dea-tools: geospatial analysis of satellite data using Digital Earth Australia, Open Data Cube, and Xarray.

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Software

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Summary

`dea-tools` is an open-source Python package for geospatial analysis of satellite data using Digital Earth Australia, Open Data Cube (ODC) and xarray (Hoyer & Joseph, 2017). It provides a broad set of utilities for loading, visualising, transforming, and analysing Earth Observation (EO) data across space and time

The package includes tools for:

- **Data handling**: Load and combine DEA data products, manage projections and resolutions.
- **Visualisation**: Create static and interactive maps, RGB plots, and animations.
- **Remote sensing indices**: Calculate band indices such as NDVI, NDWI, and more.
- **Spatial and temporal analysis**: Apply raster/vector operations, extract contours, compute temporal stats, and model change over time.
- **Machine learning and segmentation**: Train and apply classifiers, or run image segmentation workflows.
- **Parallel processing**: Set up Dask clusters for scalable processing of large datasets.
- **Domain-specific tools**: Analyse coastal change, intertidal zones, land cover, wetland and waterbody dynamics, and climate datasets.

Statement of need

Satellite remote sensing offers an unparalleled resource for

Features

Analytical tools native to Xarray

Flexible and powerful functionality for scientific analysis of xarray objects: - Extracting time series statistics, using the module `dea-tools.temporal`: * Generic summary statistics on any time series with `temporal_statistics` * Phenology with `xr_phenology` * Linear regression with `xr_regression` - Validation tools within `dea-tools.validation`: * Compute common statistical metrics with `eval_metric` * Generate random points over an xarray object with `xr_random_sampling` - Spatial analysis tools within `dea-tools.spatial`: * Vectorise xarray.DataArrays into geopandas.GeoDataFrames with `xr_vectorise` * Rasterize geopandas.GeoDataFrames into xarray.DataArrays with `xr_rasterize` * Extract subpixel contours from xarray.DataArrays with `subpixel_contours` * Interpolate point data stored in a geopandas.GeoDataFrame into an xarray.DataArray with `xr_interpolate` - Xarray wrappers for the full machine learning pipeline in in `dea-tools.classification`: * prepare data for ML predictions with `sklearn_flatten` and `sklearn_unflatten` * Fit models with `fit_xr` * ML predictions with `predict_xr` * Handle spatial autocorrelation in cross-validation workflows with `spatial_train_test_split` and `SKCV` (spatial k-fold cross validation)

Plotting

The `dea-tools.plotting` library has extensive functionality for generating beautiful imagery from xarray objects. - RGB plots for true and false colour composite images with `rgb` - Highly customisable animations with `xr_animation`

Integration with Open Data Cube libraries

dea-tools works naively with odc-geo

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References

Hoyer, S., & Joseph, H. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, *5*(1). https://doi.org/10.5334/jors.148